

1	Course Name:
	Internet of Things
2	Course Code:
	IIB48703
3	Year and Semester Offered:
	Year 2 Semester 2
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	IIB12604 Network Fundamentals
6	Course Synopsis: In this course, students will explore the three basics insights of the Internet of Things: Why do we want to connect everything? What do we want to connect? And how do we connect everything? A typical IoT solution includes sensors, local analytic abilities, network connections, and the ability to process and analyze the gathered data. Overall it is important to understand how a product or a process or a business overall can be improved with the instrumentation and the collection of data. It all starts with the end-to-end interconnection from a sensor to a gateway and from there to the network and the cloud.
7	Course Learning Outcome: Upon completion of the course, students should be able to: 1. Assembles circuits and microcontroller programs using the appropriate tools. 2. Use the appropriate programming language to program IoT systems. 3. Proposes the relevant Internet of Things solution in business communication, operation and social problems.
8	Content Outline:
	Topic
	1. Things and Connections
	2. Software is Everywhere
	3. Python Programming
	4. Sensors, Actuators, and Microcontrollers
	5. Arduino
	6. Raspberry Pi
	7. Fog Networks and Cloud Services
	8. Industrial IoT Applications
	9. Create and IoT Solution